

Such a Thing as Society

On the cost of neglect as economic policy

In 2021, the United States expanded the Child Tax Credit: a monthly payment to families with children, no strings attached. The Supplemental Poverty Measure for children dropped to 5.2%, the lowest rate ever recorded. In 2022, Congress let the expansion expire. Child poverty doubled to 12.4% in a single year, the largest increase in over fifty years of measurement.¹ Roughly 5.2 million children crossed the poverty line in one direction, then crossed it back in the other, in the time it took a provision to lapse. Not a recession. Not a structural shift. A subtraction. One dimension removed; the product collapsed.

That word, *product*, isn't a metaphor. It's multiplication. Factors, multiplied together. And when any one of them is zero, so is everything else.

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A mathematical result I've been working on, proven axiomatically and validated across eight empirical domains, shows that when independent dimensions combine to produce an emergent outcome, the composition is multiplicative, not additive.² Not a sum. A product. The proof follows from five constraints that are hard to argue with: if any dimension is absent, the outcome is zero; the function is continuous and monotonic; it's scale-consistent; and the dimensions don't interact through hidden terms. Given these, the multiplicative form is the only valid composition function. There is no additive alternative. The math looked for one. It isn't there.

The policy consequence falls straight out of the calculus. If the composition is multiplicative, the marginal return on investment in any dimension scales inversely with that dimension's current value. The lower the dimension, the higher the return. Not as a moral intuition. As a derivative. When x is near zero, the slope is nearly vertical. When x is already large, the slope is nearly flat.

The returns are at the bottom. They've always been at the bottom. The curve doesn't care what we assumed.

And if any dimension reaches zero, the product is zero. No compensation from the other dimensions. No partial credit. The system doesn't degrade gracefully; it drops off a cliff. One factor at zero, and the rest of the equation is decorative.

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Consider what this looks like at the scale of a single person.

A person without stable housing. If the ability to function composes multiplicatively from housing stability $\times$ healthcare access $\times$ education $\times$ financial security $\times$ social connection, then a person with zero housing gets zero return from job training. Not diminished returns. Not inefficient returns. Zero. The job training doesn't land because the person can't sleep, can't shower, can't keep a schedule, can't store documents, can't show up consistently enough for any other investment to compound. Everything else you pour in multiplies by zero. It isn't waste in the usual sense. It's worse than waste. It's investment that was structurally guaranteed to fail before it started.

This isn't theory. It's what Housing First programs discovered by accident, and then confirmed with data that leaves no room. The CDC reviewed 26 studies: Housing First decreases homelessness, increases stability, improves quality of life; every dollar invested returns \$1.44 in cost savings elsewhere.³ The Canadian At Home/Chez Soi trial, a randomized controlled study across five cities with 950 participants, found 73% stable housing at one year versus 31% for treatment-as-usual; 69% of the intervention's cost was offset by reductions in emergency services, shelters, and hospitals that were no longer needed.⁴ Finland adopted Housing First as national policy in 2008. Long-term homelessness fell 68% over fourteen years. Finland is now the only country in the European Union where homelessness has declined.⁵ The only one.

The conventional approach, the one most countries still use (job training here, a clinic visit there, a food bank on Thursdays), is implicitly additive. It assumes each intervention adds independent value. The multiplicative result says that assumption is structurally wrong. If housing is zero, the food bank adds to zero. If healthcare access is zero, the job training multiplies by zero. You are investing in dimensions that cannot produce returns because the product is already collapsed by the one dimension you're not touching.

Housing First works not because housing is the most important dimension. It works because housing was the zero. Until you repair the zero, nothing else compounds. The piecemeal model keeps pouring resources into nonzero dimensions while the zero swallows the product whole.

The "bootstrap" narrative (work harder, try more, pull yourself up) is not merely cruel. It is mathematically incoherent when any dimension is at zero. Effort in the nonzero dimensions

multiplies by zero. The math doesn't have a bootstrap exception. Neither does the data.

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Now scale up.

If a country's economic resilience composes multiplicatively from safety nets $\times$ healthcare access $\times$ employment $\times$ infrastructure $\times$ institutional trust, then during a crisis, the dimensions nearest zero produce the highest marginal returns per dollar invested. The mathematically optimal crisis response is to invest in the lowest dimensions. Not because it's compassionate. Because it's where the slope is steepest. The math is indifferent to your ideology. It just tells you where the leverage is.

Austerity inverts this. It cuts the dimensions closest to zero (social programs, unemployment support, healthcare) while protecting the dimensions already above zero: bank recapitalization, corporate liquidity, bond market confidence. In multiplicative terms, austerity is a policy of investing where the derivative is flattest and divesting from where it's steepest. It is, by construction, the least efficient allocation available.

Greece and Iceland are the natural experiment, and the results are not subtle. Both economies collapsed in 2008. Greece chose austerity. GDP fell 26% in five years. Unemployment peaked at 28%. Youth unemployment hit 58%. The economy shrank by a quarter, and the debt-to-GDP ratio, the thing austerity was supposed to fix, climbed from 130% to 180%.⁶ Austerity failed to achieve even its own stated objective.

Iceland chose the opposite: let the banks fail, protect domestic deposits, prosecute the bankers (29 of them went to prison), invest in the safety net while the currency devalued. GDP returned to pre-crisis levels by 2015. Unemployment fell to 3%. IMF loans were repaid early.⁷ Same crisis. Opposite policy. Opposite result. The one that protected the floor recovered. The one that cut the floor did not.

The United States and the United Kingdom ran the same experiment one year later. The US passed the American Recovery and Reinvestment Act: \$836 billion targeting unemployment compensation, state fiscal aid, infrastructure, food assistance. The CBO estimated it raised real GDP by 0.7–4.1% and added up to 2.8 million jobs.⁸ The UK chose austerity under Cameron and Osborne: capital spending fell 32% in real terms; food bank centres went from 35 to nearly 1,300; emergency food parcels from 60,000 to 2.89 million; homelessness rose 34%; life expectancy flatlined. After fourteen years of cutting, government debt rose from 65% to 98% of GDP.⁹ Austerity did not even reduce the debt. It grew the debt and damaged everything else.

The IMF, not a radical institution, admitted the error. Blanchard and Leigh found the actual fiscal multiplier was 0.9 to 1.7, not the 0.5 their models had assumed.¹⁰ Cutting the lowest dimensions cost roughly twice what the forecasters thought it would. The models were wrong because they assumed additive composition. The world is multiplicative.

The Keynesian result (stimulus outperforms austerity in downturns) falls out of the multiplicative math without needing macroeconomic theory at all. Stimulus invests where the derivative is steepest. Austerity cuts there. That's all there is to it. The rest is accounting.

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Scale up further, to the level of a whole society, and the prediction gets uglier.

If a society's output composes multiplicatively from institutional trust $\times$ human capital $\times$ infrastructure $\times$ financial stability, then one dimension reaching zero doesn't produce a proportional decline. It produces a cliff. The product doesn't thin out. It disappears.

Venezuela is sitting on 303 billion barrels of proven oil reserves. That's roughly 18% of the world's total. More than Saudi Arabia.¹¹ It has labor, infrastructure, arable land. None of it mattered. GDP peaked at \$372.6 billion in 2012. By 2020, it had fallen 88%, deeper than the US Great Depression, worse than Syria's wartime collapse. Per capita income dropped from \$12,607 to \$1,509. The oil never moved. The reserves didn't shrink. What collapsed was institutional trust: the World Bank's Rule of Law estimate fell from -0.76 to -2.35 ; Transparency International's Corruption Perceptions Index dropped to 10 out of 100.¹² One dimension went to zero, and three hundred billion barrels of oil became irrelevant. The product doesn't care how large your other factors are.

Zimbabwe had productive farmland and an educated workforce. After the land-reform seizures beginning in 2000, food production fell 60% in a decade. Hyperinflation peaked at 79.6 billion percent per month. Prices doubled every twenty-four hours. Per capita income fell from \$1,640 to \$661, and by 2008 GDP per capita had collapsed to a level last seen in 1952. The World Bank's Rule of Law indicator hit the 1st percentile.¹³

Look at the arithmetic. In both cases, large dimensions survived: oil, farmland, human capital, physical infrastructure. One dimension collapsed. An additive model predicts that losing one of four dimensions should reduce output by about a quarter. Actual decline: 78–90%. That gap between 25% and 90% is the difference between addition and multiplication. It is the cliff. And it is exactly what the multiplicative form predicts: one factor at zero, and the product goes with it. Everything else is decoration.

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This is where the conventional wisdom breaks.

For forty years, the dominant economic policy framework in the United States and much of the West has been: invest at the top. Cut taxes for the wealthy and for corporations; the benefits trickle down. The implicit model is additive. Boost the top, the gains distribute, everyone's sum increases. It's a clean story. It's also wrong, and we can now prove it's wrong.

The multiplicative result says trickle-down invests where the derivative is flattest. When x is already large, the slope is nearly horizontal. You're pouring money onto a plateau and waiting for it to run downhill. It doesn't. The returns at the top are, within measurement error, zero.

And the data is not ambiguous about this.

Hope and Limberg examined fifty years of major tax cuts for the rich across eighteen OECD countries, 1965 to 2015, and found no statistically significant effect on GDP per capita or unemployment. None. The only measurable outcome was higher income concentration at the top.¹⁴ The Congressional Research Service examined sixty-five years of top-rate reductions and found no association with saving, investment, or productivity growth; what the cuts *did* track with was the top 0.1% income share rising from 4.2% to 12.3%.¹⁵ That report was withdrawn under Senate Republican pressure. It was reissued, substantively unchanged. The data was too clean to revise. The IMF concluded that the equity-efficiency trade-off is "largely illusory": lower inequality is robustly correlated with faster, more durable growth.¹⁶ Kansas ran what its own governor called a "real live experiment": cut the top rate 30%, tax pass-through income at zero. GDP grew at half the national rate. The experiment was so conclusive that a Republican legislature repealed the cuts over the governor's veto.¹⁷

Now look at the bottom.

In rural Kenya, unconditional cash transfers of about \$1,000 to over 10,500 poor households produced a local fiscal multiplier of 2.5. Each dollar at the bottom generated \$2.50 in local economic activity, with large positive spillovers to non-recipients and minimal inflation.¹⁸ The BRAC graduation program, a simultaneous push across multiple dimensions (asset transfer, consumption support, training, coaching, savings, health), tested across six countries in a randomized trial, produced returns of 133% to 433%. The multidimensional package outperformed single-component interventions in the same study.¹⁹ That distinction matters: it's the multiplicative approach beating the additive approach in a head-to-head trial. Chetty, Hendren, and Katz found that children who moved to lower-poverty neighborhoods before age thirteen earned 31% more as adults. Returns were largest for those starting from the lowest

baseline.²⁰ The steepest part of the curve. Every time.

Place these next to each other. Investment at the top: zero measurable growth across fifty years and eighteen countries. Investment at the bottom: multipliers of 2.5x to 4.3x in randomized, controlled trials. Fifty years of data on one side. Six RCTs on the other. Trickle-down invests where the derivative is flat. Social investment invests where the derivative is vertical. One is provably suboptimal. The other is provably optimal. This is not a close call. The math closed the question.

What remains open is why we keep choosing the answer that doesn't work.

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On July 4, 2025, the United States signed into law the One Big Beautiful Bill Act. It cuts roughly \$1 trillion from Medicaid and \$186 billion from SNAP over ten years. Those are the two largest safety net programs in the country. They serve the lowest dimensions of the most vulnerable populations. The same law makes permanent the 2017 tax cuts, raises the estate tax exemption to \$15 million, and restores full bonus depreciation for corporations. The CBO's distributional analysis says it plainly: the bottom income decile loses 3.1% of its resources; the top decile gains 2.7%.²¹

Read that again. The floor gets cut. The ceiling gets subsidized. The gap between them doesn't widen as a side effect. Widening the gap is the mechanism.

The cliff indicators are not hypothetical. They're already in the data. Homelessness hit a record 771,480 in HUD's 2024 count: an 18% increase, the largest jump since they started counting in 2007.²² In November 2025, SNAP funding lapsed for the first time in the program's history, leaving 42 million people without food assistance. Food banks hit record demand.²³ Curtis Medical Center in Nebraska became the first rural hospital to attribute its closure directly to anticipated Medicaid cuts. Over 300 more are on the list.²⁴

And the institutional trust dimension, the one that in Venezuela and Zimbabwe swallowed the oil and the farmland and the educated workforce and everything else, is drifting downward. The V-Dem Institute downgraded the United States from "liberal democracy" to "electoral democracy" for the first time in over fifty years.²⁵ Freedom House gave it the lowest score since the current system began.²⁶ The Economist Intelligence Unit recorded its lowest US score since the index was created in 2006.²⁷ Gallup finds trust in the federal government near five-decade lows.²⁸

None of these is collapse. But the multiplicative model doesn't wait for every dimension to collapse. It needs one.

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The math doesn't have a political opinion. It has a structure. The structure says the returns are at the bottom, not because the bottom deserves them, not because compassion demands it, but because that's where the curve is steepest. Helping the most affected is not charity. It is optimization. The compassionate allocation and the efficient allocation turn out to be the same allocation. They were never in tension. We just had the wrong model.

For fifty years, the model has been additive: invest at the top, the gains distribute. The multiplicative result says that model is structurally wrong. And the evidence, across eighteen countries, six randomized trials, two failed states, and two natural experiments in crisis response, says the same thing the math says. The returns at the top are, within measurement error, zero. The returns at the bottom are 2.5 to 4.3 times the investment. The question of where to invest was never a values question. It was a math question, and it's been answered.

Why we keep choosing the answer that doesn't work is a different kind of question, one the math can't reach. A system that profits from investing in the wrong place has structural reasons to continue, and structural machinery for making sure the people who bear the cost blame each other instead of the allocation. That isn't mathematics. That's politics. I'll leave it there.

Thatcher said there was no such thing as society. The math says there is. It says society is a product, not a sum. It says the product is only as strong as its weakest factor. And it says the returns, if anyone is interested in collecting them, are at the bottom.

They've been there the whole time.

Sources

¹ U.S. Census Bureau, Supplemental Poverty Measure, 2022. Columbia Center on Poverty and Social Policy. CBPP: "the largest on record."

² See davidkirsch.me/research. Axiomatic uniqueness proof following Aczél (1966). Validated across eight domains. Repository: github.com/duke-of-beans/multiplicative-composition.

³ CDC Community Preventive Services Task Force, systematic review of 26 Housing First studies. NLIHC, 2023.

⁴ Aubry et al. *Psychiatric Services*, 2020. DOI: 10.1176/appi.ps.202000029. N=950, five cities, 2009–2011.

⁵ Pathfinders, 2024. Finnish government: long-term homelessness reduced 68%, 2008–2022. €270M invested.

⁶ Eurostat; Danchev, *Fiscal Studies*, 2024; Peterson Institute; Oxfam.

⁷ Wharton, 2018; Seven Pillars Institute; Oxford Academic, 2022.

⁸ CBO publications 25005, 25075, 49958. Final budgetary impact: \$836–840B.

- ⁹ Institute for Government; Trussell Trust / House of Commons Library, May 2025; Bramley et al., *BMC Public Health*, 2022.
- ¹⁰ Blanchard & Leigh, IMF Working Paper 13/1, January 2013.
- ¹¹ OPEC Annual Statistical Bulletin 2025. ~303B barrels proven reserves.
- ¹² World Bank WGI; Transparency International CPI; IMF; Institute of International Finance.
- ¹³ World Bank WGI; Hanke & Kwok (hyperinflation).
- ¹⁴ Hope & Limberg, *Socio-Economic Review* 20(2):539–559, 2022. DOI: 10.1093/ser/mwab061.
- ¹⁵ Hungerford, CRS Report R42729, September 2012. Withdrawn under Senate Republican pressure; reissued unchanged.
- ¹⁶ Ostry, Berg & Tsangarides, IMF Staff Discussion Note 14/02, February 2014.
- ¹⁷ Kansas tax cuts repealed June 6, 2017. Gale, Brookings; CBPP; Tax Policy Center.
- ¹⁸ Egger et al., *Econometrica* 90(6):2603–2643, 2022. DOI: 10.3982/ECTA17945.
- ¹⁹ Banerjee, Duflo et al., *Science* 348(6236):1260799, 2015. Returns: 133% (Ghana) to 433% (India).
- ²⁰ Chetty, Hendren & Katz, *AER* 106(4):855–902, 2016. DOI: 10.1257/aer.20150572.
- ²¹ CBO distributional analysis, OBBBA (P.L. 119-21), August 11, 2025. Tax Policy Center; ITEP.
- ²² HUD, 2024 Annual Homelessness Assessment Report. PIT count: 771,480.
- ²³ NPR, Deseret News, CNBC, November 2025. CBPP SNAP Tracker.
- ²⁴ Curtis Medical Center: Nebraska Public Media, 2025. UNC Sheps Center: 300+ at risk.
- ²⁵ V-Dem Institute, *Democratization* Vol. 32, 2025.
- ²⁶ Freedom House, *Freedom in the World* 2026. US: 81/100.
- ²⁷ EIU, *Democracy Index* 2025. US: 7.65, 34th globally.
- ²⁸ Gallup, September 2025. Trust in federal government: 38%.